

Mohammad Golam Sarowar



📍 Khilkhet, Dhaka, Bangladesh

✉ mohammad.sarowar06@gmail.com ☎ +8801876473956

Distributed Systems | IoT | Data Science | AI/ML | Federated Learning | Network Security

RESEARCH INTEREST

My research interests focus on distributed and intelligent computing, with particular interest in Data Science, AI/ML, IoT, and Federated Learning for developing secure, privacy-aware, and reliable digital systems. My background in computer networking and network security further supports my interest in resilient distributed architectures and intelligent networked systems.

EDUCATIONAL BACKGROUND

Bachelor of Science in Computer Science and Engineering | Project-Based | 2024

Department of Computer Science and Engineering | Bangladesh University | Dhaka, Bangladesh.

Project Title: *Design, Implementation, and Performance Analysis of Linux-Based Networking Using Ubuntu Server*

Result/CGPA: 3.64/4.00

Grading Scale: 80% and above = 4.00; 75–<80% = 3.75; 70–<75% = 3.50 on a 4.00-point scale.

Diploma in Engineering in Computer Technology | 4-Year Program | Examination 2016

| Result Published 2017 | Feni Polytechnic Institute | Bangladesh Technical Education Board (BTEB) | Feni, Bangladesh

Result/CGPA: 3.38/4.00 | Academic Rank: 3rd out of 40

Industrial Training: 6 Credit Hours | Grade: A+ (4.00/4.00)

Grading Scale: 80% and above = 4.00; 75–<80% = 3.75; 70–<75% = 3.50 on a 4.00-point scale.

Dakhil (Secondary School Certificate Equivalent) | 2012

Group: General | Hazrat Shah Sufi Main Uddin Shah (R.) Dakhil Madrasah | Chattogram, Bangladesh

Board: Bangladesh Madrasah Education Board, Dhaka

Result/GPA: 4.81/5.00 | Academic Rank: 2nd out of 36

Grading Scale: 80–100% = 5.00; 70–79% = 4.00; 60–69% = 3.50 on a 5.00-point scale.

HONORS AND AWARDS

Early Career Promotion– Assistant Manager | Digi Jadoo Broadband Limited | 2026

Recognized with an early promotion to Assistant Manager for outstanding performance, technical expertise, and professional contribution.

Best Employee Award | Digi Jadoo Broadband Limited | 2025

Recognized for outstanding professional performance and contribution to the organization

3rd Highest Academic Result – Final-Year Cohort | Department of Computer Science and Engineering (CSE) | Bangladesh University | 2024

Ranked 3rd in academic performance among the final-year students in the Department of Computer Science and Engineering.

Highest CGPA Recognition – 5th Semester | Bangladesh University | 2022

Recognized for achieving the highest academic performance in the 5th semester of the undergraduate program.

Merit Scholarship | Diploma in Engineering | Feni Polytechnic Institute

Received a merit-based scholarship in recognition of academic performance.

Best Performer – Networking Training | Bijoy Online Limited | 2018

Achieved the highest score in the final training assessment among 7 participants.

Academic Merit – 2nd Position | Dakhil Examination | 2012

Achieved 2nd position based on academic performance among the 2012 batch at the institution.

TEACHING AND MENTORING

Peer Tutor & Study Group Leader | Bangladesh University | 2021-2024

- Provided academic support to approximately 15–20 undergraduate students
- Organized group study and discussion sessions, explained difficult concepts, and assisted classmates with problem-solving and examination preparation.

Student & Academic Mentor | Bangladesh University | 2021-2024

- Guided university applicants, junior students, and peers on academic planning, study strategies, university admission, higher education, and scholarship opportunities.
- Provided guidance on career planning, technical skill development, professional certifications, and networking careers.

Course Representative | Bangladesh University | Under the Guidance of Muhammad Minoar Hossain | 3rd Semester | 2022

- Served as the Course Representative for Data Structure, acting as a communication link between the course instructor and students.
- Coordinated course-related announcements, academic updates, and communication among classmates.
- Assisted students with course-related concerns and helped maintain effective coordination between the instructor and the class.

Private Tutor | Feni Polytechnic Institute | 2014–2016

- Provided private tutoring to approximately 8–10 diploma-level students in Computer Fundamentals, Programming, and Mathematics.
- Delivered personalized one-to-one and small-group academic support focused on concept development, problem-solving, and examination preparation.

WORK EXPERIENCE

Assistant Manager – Core IP Network | Digi Jadoo Broadband Ltd. | Dhaka, Bangladesh | Jul 2023–Present

Industry/Sector: Telecommunications / Internet Service Provider (ISP)

- Manage and optimize Core IP network infrastructure, ensuring network availability, scalability, performance, and reliability.
- Configure, monitor, and troubleshoot BGP, OSPF, MPLS, IPv4/IPv6, routing, switching, and traffic-engineering solutions across ISP backbone networks.
- Analyse network traffic, routing behaviour, bandwidth utilization, and network performance to identify and resolve operational issues.
- Manage BGP routing policies, peering, IP transit, and CDN traffic, coordinating with upstream providers and network partners.
- Perform network monitoring, incident troubleshooting, root-cause analysis, capacity planning, and infrastructure upgrades.
- Contribute to Core Network Expansion, BGP Traffic Engineering, CDN Integration & Optimization, IPv6 Deployment, and ISP Backbone Upgrades.

Senior Engineer – Technical Department | Link3 Technologies Ltd. | Dhaka, Bangladesh | Mar 2022 – Jun 2023

Telecommunications / Internet Service Provider (ISP)

- Managed and maintained ISP network infrastructure to support stable, reliable, and high-availability connectivity.
- Configured and troubleshoot BGP, OSPF, VLANs, PPPoE, routing, switching, and IPv4/IPv6 networks.
- Performed network monitoring, fault analysis, and incident resolution across Cisco, Juniper, Huawei, and MikroTik platforms.
- Coordinated with NOC teams, upstream providers, and technical teams for network troubleshooting, maintenance, and service restoration.
- Maintained network configurations and technical documentation while supporting day-to-day ISP network operations.

CERTIFICATIONS & PROFESSIONAL TRAINING

Professional Certifications

- **Cisco Certified Network Associate (CCNA)** | Cisco | Feb 2024–Feb 2027
Cisco ID: CSC014551272
- **MikroTik Certified Routing Engineer (MTCRE)** | MikroTik | Jan 2024
Certificate ID: 2401RE4147
- **MikroTik Certified Network Associate (MTCNA)** | MikroTik | Oct 2023
Certificate ID: 2310NA6582

Professional Training

- **ISP/IIG Setup using Cisco (IOS-XR)** | Open ICT | Jun 2026 (*Certificate of Attendance*)
- **ISP/IIG Setup using Juniper** | Open ICT | Oct 2024 (*Certificate of Attendance*)
- **Basic & Advanced Networking Training** | Bijoy Online Limited | 2018
3-month practical training

RESEARCH EXPERIENCE

2026–Ongoing | 2nd Author | **Distributed Telemedicine Systems in Bangladesh: A Review of AI, Federated Learning, IoMT, Blockchain, and Edge Computing for Secure Rural Healthcare (Probable title)**

Under the supervision of Rafat Ara, Assistant Professor & PhD Researcher, Department of Computer Science and Engineering, University of Information Technology & Sciences (UITS), Bangladesh

- Conducting a comprehensive review of **distributed telemedicine systems**, focusing on the integration of AI, Federated Learning, IoMT, Blockchain, and Edge Computing.
- Investigating **security, privacy, accessibility, and technological challenges** in distributed healthcare systems, with particular emphasis on rural healthcare in Bangladesh.
- Reviewing existing research to identify **technology gaps, challenges, and potential future research directions** for secure and scalable digital healthcare.

2026 | **Researcher & Co-Author | IoT-Based Smart Contract Framework for Rice Supply Chain Traceability Recall System and Consumer Safety**

International Conference on Technological Intelligence and Business Strategies (TIBS 2026), Manila, Philippines

- Contributed to research on an **IoT and smart contract-based framework** for rice supply-chain traceability, product recall, and consumer safety.
- Explored the application of **IoT and blockchain-based smart contracts** to improve transparency, traceability, and reliability in the agricultural supply chain.
- Paper selected for publication in connection with **TIBS 2026**, subject to fulfilment of Springer publication guidelines.

PROJECTS & PRACTICAL EXPERIENCE

2024 | **Enterprise Linux Server Infrastructure Deployment**

Academic Project | Bangladesh University

- Designed and deployed a Linux-based server infrastructure using Ubuntu Server.
- Configured core networking services and evaluated server performance in a practical networking environment.

- Applied practical knowledge of Linux administration, networking, and server deployment in an academic project environment.

2023–Present | ISP Core Network Optimization

Professional Technical Project | Digi Jadoo Broadband Ltd.

- Contributed to Core IP network optimization through BGP traffic engineering, routing optimization, and peering coordination.
- Supported IPv4/IPv6 deployment, CDN integration, and backbone performance improvement initiatives.

2022–2023 | Network Monitoring & Infrastructure Support

Professional Technical Project | Link3 Technologies Ltd.

- Supported network monitoring, troubleshooting, configuration management, and infrastructure maintenance activities.
- Contributed to maintaining network availability and resolving operational issues across ISP network infrastructure.

RESEARCH SKILLS

Research & Academic Skills

Literature Review | Literature Search & Information Collection | Critical Reading | Academic Writing | Referencing & Citation | Technical Documentation

Technical & Analytical Skills

Computer Networking | Network Troubleshooting | Network Monitoring | Basic Data Analysis | IoT Fundamentals | Network Security Fundamentals

Programming & Data Tools

Python (Basic) | C (Basic) | Microsoft Excel | Microsoft Word | GitHub

Research & Visualization Tools

Google Scholar – Literature Search | **Mendeley** – Reference Management | **draw.io** – Diagrams & System Architecture

OS & TECHNICAL TOOLS

Operating Systems:

Linux (Ubuntu, CentOS) | Windows

Network Operating Systems & Platforms:

Cisco IOS / IOS-XR | Junos OS | Huawei VRP | MikroTik RouterOS / Winbox

Network Monitoring & Analysis:

PRTG | Zabbix | Cacti | Wireshark

Virtualization Platforms:

VMware | Proxmox

Remote Access & Administration:

SecureCRT | PuTTY

WORKSHOPS, TRAINING & PROFESSIONAL DEVELOPMENT

15-Day Intensive Workshop on “Targeting Prestigious Erasmus Mundus Scholarship”

ABCD Laboratory, Bangladesh | 2026

Under the Guidance of: Md. Junaid, Founder & Principal Investigator, ABCD Laboratory, Bangladesh

- Participated in an intensive workshop focused on Erasmus Mundus scholarship preparation, higher-study planning, and academic profile development.

Aspire Leaders Program – Cohort 3, 2026

Aspire Institute | July 2026 – Present

Selected Participant

- Selected for an 8-week global leadership development program focused on leadership, communication, career readiness, mentorship, and global peer networking.

Basic & Advanced Networking Training

Bijoy Online Limited | 2018

- Completed a 3-month practical networking training covering WAN/LAN configuration, routing, VLANs, firewall, VPN, OSPF, BGP, bandwidth management, and network administration.

ACADEMIC CONFERENCES & PRESENTATIONS

International Conference on Technological Intelligence and Business Strategies (TIBS 2026)

Manila, Philippines | Online Presentation | 2026

Paper: *IoT-Based Smart Contract Framework for Rice Supply Chain Traceability Recall System and Consumer Safety*

Role: Co-Author & Conference Presenter

- Presented the research paper at TIBS 2026 through an online conference presentation.
- Discussed an **IoT- and smart-contract-based framework** for improving supply-chain traceability, product recall, data integrity, and consumer safety.

Academic Presentations – Undergraduate Studies

Bangladesh University | 2021–2024

- Delivered individual and group academic presentations on technical and course-related topics as part of undergraduate CSE coursework.

SELF-STUDY & ONLINE LEARNING

Networking & Infrastructure

- **ISP/IIG Networking & Core Network Technologies** — Self-directed study through online technical resources and practical exploration.
- **BGP, OSPF, MPLS & IPv6** — Continuous self-learning focused on routing, traffic engineering, network scalability, and modern IP networking.
- **Network Security & Traffic Analysis** — Self-study of network security concepts, traffic analysis, and practical troubleshooting approaches.

AI, Data & Emerging Technologies

- **Python & Data Analysis** — Online learning and hands-on practice with Python for basic data processing and analysis.
- **Artificial Intelligence & Machine Learning** — Self-study of fundamental AI/ML concepts and their applications in computing systems.
- **Federated Learning** — Independent learning focused on decentralized and privacy-preserving machine learning, particularly for healthcare applications.
- **IoT & Blockchain** — Self-study of IoT architectures, smart contracts, and blockchain-based applications relevant to secure distributed systems.

Research & Academic Development

- **Research Methodology** — Self-directed learning through academic resources and research literature.
- **Scientific Writing & Literature Review** — Developing skills in academic writing, literature analysis, citation, and manuscript preparation.
- **Research Publication Process** — Learning about manuscript preparation, conference presentation, peer-reviewed publication, and academic research practices.

LEADERSHIP EXPERIENCE

Assistant Class Representative (ACR) | Bangladesh University | 2021–2024

- Elected by a majority vote of classmates to serve as **Assistant Class Representative**.
- Facilitated communication between students and faculty by sharing academic updates and conveying students' feedback and concerns.
- Supported academic coordination and helped maintain effective communication within the class.

Project Leader – Linux-Based System Administration & Network Infrastructure Project | Bangladesh University | 2024

Under the Supervision of Dr. Uzzal Prodhan, Professor, Department of CSE

- Led project planning, technical implementation, troubleshooting, and documentation activities.

- Coordinated team members throughout the project and supported effective task execution.
- Worked on Linux environment configuration, system management, security practices, and network-related tasks.
- Applied practical knowledge of Linux administration and networking in an academic-industry collaborative project.

Research Coordinator | Collaborative Research Team | 2026–Present

- Coordinated research activities related to emerging technologies and supported research development.
- Contributed to literature review, technical discussions, research planning, and preparation of research outcomes.
- Collaborated with research team members to support manuscript and research development activities

PROFESSIONAL LEADERSHIP

Team Leader – Network Operations

Digi Jadoo Broadband Ltd. | Current

- Lead Network Operations activities and coordinate day-to-day NOC operations.
- Supervise team members and provide technical guidance on network operations and troubleshooting.
- Coordinate with clients, vendors, and relevant technical teams to resolve critical network issues.
- Support reliable and continuous network performance through effective operational coordination.

VOLUNTEERING & COMMUNITY ENGAGEMENT

Volunteer Member | Bangladesh Red Crescent Society (RCY)

Bangladesh | 2021–Present

- Participated in community service, educational support, and social welfare activities through the Bangladesh Red Crescent Youth (RCY).
- Supported community-oriented initiatives and developed experience in teamwork, communication, and social engagement.

Community Volunteer – Blood Donation Activities

Bangladesh | 2018–Present

- Participated in voluntary blood donation initiatives and supported emergency blood-donor coordination.

- Contributed to community awareness activities related to voluntary blood donation and social support.

Educational Volunteer & Peer Support

Bangladesh University | 2021–2024

- Supported junior students and peers with academic guidance, programming and networking-related assistance.
- Shared information on higher education, scholarship opportunities, research, and professional development.

ACADEMIC & INTERNATIONAL EVENTS

64th Bangladesh International Education Expo 2026

International Convention City Bashundhara (ICCB), Dhaka | 23–25 July 2026

Participant

- Participated in an international education event focused on higher education opportunities, scholarships, and international university programs.
- Explored opportunities related to study abroad, scholarship pathways, and international academic collaboration.

EXTRACURRICULAR ACTIVITIES

Badminton Winner – Office Tour

Bijoy Online Limited | 2018

- Won the Badminton competition during an organizational office tour.

School-Level Sports Activities

2007–2012

- Participated in various school-level sports and recreational competitions, including running and team-based games, and received prizes for performance.

PROFESSIONAL & ACADEMIC MEMBERSHIPS

- **International Association of Engineers (IAENG) | Member** | Member No. 574155 | Since July 2026
- **Institute of Research Engineers and Doctors (IRED) | Member**
- **CSE Club, Bangladesh University | Former Member**

LANGUAGE SKILLS

- **Bengali:** Native proficiency
- **English:** Advanced proficiency
- **IELTS:** Currently preparing; target test date 2026

REFERENCES

Dr. Uzzal Prodhhan

Professor, Department of Computer Science & Engineering
Bangladesh University, Bangladesh

Email: uzzal.prodhhan@bu.edu.bd

Phone: +8801718-045304

Relationship: Undergraduate Project Supervisor & Research Supervisor

Rafat Ara

Assistant Professor, Department of Computer Science & Engineering
University of Information Technology & Sciences (UITS), Bangladesh

Email: rafat.ara@uits.edu.bd

Phone: +8801677061760

Relationship: Research Mentor

Tonmoy Mutsuddy

Senior Manager, ISP Support & NOC
Digi Jadoo Broadband Ltd., Bangladesh

Email: tonmoy@digijadoo.com

Phone: +8801709634256

Relationship: Immediate Supervisor